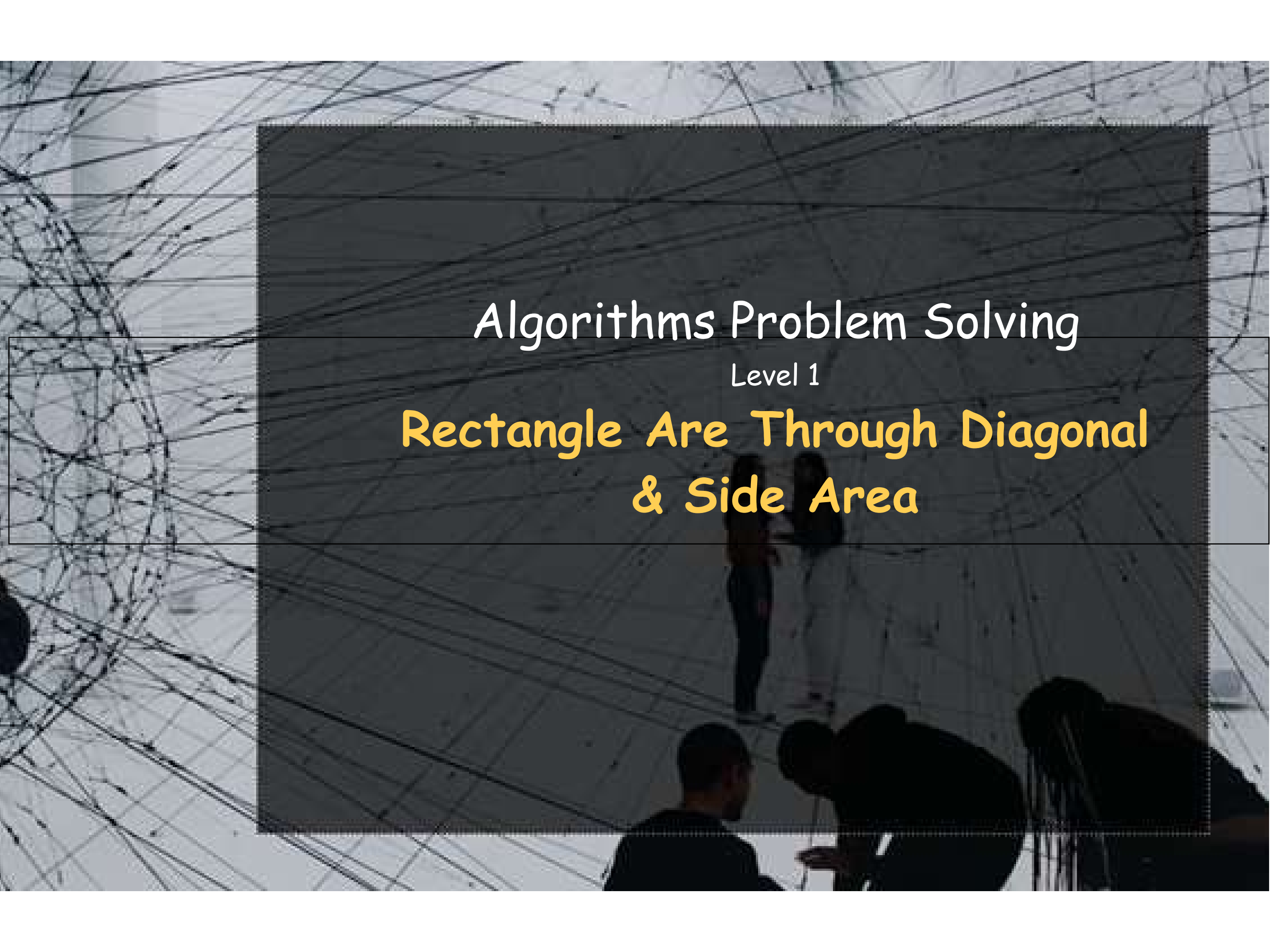


Algorithms Problem Solving

Level 1

Rectangle Area Through
Diagonal & Side Area

The background of the slide features a whiteboard with a grid pattern. In the foreground, the silhouettes of several people are visible, appearing to be in a meeting or collaborative work environment. The text is overlaid on a dark rectangular area in the center of the whiteboard.

Algorithms Problem Solving

Level 1

Rectangle Area Through Diagonal & Side Area

Definition

- ◆ This program calculates the area of a rectangle when one side and the diagonal are given.
- ◆ The program uses the Pythagorean theorem, `sqrt()`, `pow()`, multiplication, functions, and float values.
- ◆ The area is calculated by finding the missing side first, then multiplying it by the given side.

Explanation

- ◆ The readNumbers() function reads the rectangle's side A and diagonal D .
- ◆ The RectangleArea() function calculates the missing side using the Pythagorean theorem:
 - ◆ Missing side = $\sqrt{D^2 - A^2}$
 - ◆ The rectangle area is then:
 - ◆ Area = $A \times \sqrt{D^2 - A^2}$
 - ◆ pow(D , 2) calculates D^2 .
 - ◆ pow(A , 2) calculates A^2 .
 - ◆ sqrt() calculates the square root.
- ◆ The function returns the area as a float because the result can contain decimal values.
- ◆ The printRectangleArea() function calls RectangleArea() and prints the result.

Steps in main()

1. Read the rectangle side A .
2. Read the rectangle diagonal D .
3. Calculate $D^2 - A^2$.
4. Calculate the square root of the result.
5. Multiply the missing side by A .
6. Return the calculated area.
7. Print the rectangle area.

💡 Example

◆ If the user enters:

Side: 5

Diagonal: 40

◆ First, calculate the missing side:

$$\text{Missing side} = \sqrt{40^2 - 5^2}$$

◆ Then calculate the area:

$$\text{Area} = 5 \times \sqrt{1600 - 25}$$

◆ The result is approximately 198.431.

➡ Result

Rectangle Area = 198.431

Note

- ◆ The Pythagorean theorem is used because the diagonal and the two sides form a right triangle.
- ◆ The diagonal must be greater than the given side for the square root calculation to produce a valid real number.

⊘ Warning

⚡ If D is less than or equal to A , the expression $D^2 - A^2$ will not produce a positive value required for a valid rectangle with these dimensions.



Comparison

Given Side A

- ✓ One side of the rectangle is known.
- ✓ It is multiplied by the calculated missing side.
- ✓ It is used directly in the area formula.

Diagonal D

- ✓ The diagonal is not multiplied directly to calculate the area.
- ✓ It is used with A to find the missing side.
- ✓ Its square is used in the Pythagorean calculation.

✦ Summary

- ✓ The Pythagorean theorem can be used to find the missing side of a rectangle when the diagonal and one side are known.
- ✓ `pow()` can calculate a number raised to a power.
- ✓ `sqrt()` calculates the square root.
- ✓ `float` is used because the calculated area can contain decimal values.
- ✓ The rectangle area is calculated by multiplying the two sides.

Thank you 😊